THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 14: **Probability** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The probability of a sure event is: 1

(a) 0 **(b)** 1 **(c)** $\frac{1}{2}$ **(d)** 100

2. If $P(E) = \frac{2}{3}$, then $P(\text{not } E)$ is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** 1 **(d)** 0

3. The probability of getting an odd number when a die is thrown is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{2}{3}$

4. The probability of drawing a red king from a deck of 52 cards is: 1

(a) $\frac{1}{26}$ **(b)** $\frac{1}{13}$ **(c)** $\frac{1}{52}$ **(d)** $\frac{2}{13}$

5. A student states: "If $P(A) = 0.7$, then $P(\text{not } A) = 0.7$." The correct response is: 1

(a) Correct **(b)** Incorrect; $P(\text{not } A) = 0.3$

(c) Correct only if A is sure **(d)** Cannot be determined

6. A coin is tossed twice. The probability of getting exactly one head is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{3}{4}$ **(d)** 1

7. A bag has 4 red and 6 blue balls. Student A says P(red) $= \frac{4}{10}$; Student B says P(red) $= \frac{4}{6}$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

8. Which of the following can be the probability of an event? 1

(a) 1.01 **(b)** −0.5 **(c)** $\frac{3}{2}$ **(d)** 0.99

9. A letter is chosen at random from the word MATHEMATICS. The probability that it is M is: 1

(a) $\frac{2}{11}$ **(b)** $\frac{1}{11}$ **(c)** $\frac{3}{11}$ **(d)** $\frac{1}{2}$

10. A die is thrown once. The probability of getting a number less than 7 is: 1

(a) 1 **(b)** 0 **(c)** $\frac{5}{6}$ **(d)** $\frac{1}{6}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A bag contains 5 red and 3 green balls. One ball is drawn at random. Find the probability that it is 2

(i) green

(ii) not green.

12. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting 2

(i) a black face card

(ii) the jack of hearts.

13. A student says: "It will either rain tomorrow or it will not, so the probability of rain tomorrow is $\frac{1}{2}$." Justify or refute. 2

14. Harpreet tosses two different coins simultaneously. Find the probability that she gets 2

(i) at least one tail

(ii) two heads.

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two dice are thrown simultaneously. Find the probability that **3**

(i) the product of the numbers is 6

(ii) the sum is 10

(iii) both numbers are odd.

16. A box contains 100 cards numbered 2 to 101. One card is drawn at random. Find the probability that the number is **3**

(i) even

(ii) a perfect square

(iii) a prime less than 20.

17. In a lottery there are 10 prizes and 25 blanks. Find the probability of **3**

(i) getting a prize

(ii) getting a blank.

(iii) If 5 more prizes are added, find the new probability of getting a prize.

OR

A bag contains white, black and red balls, 30 in all. The probability of drawing a white ball is $\frac{1}{5}$ and of a black ball is $\frac{1}{2}$. Find the number of red balls. **3**

18. A child has a die whose six faces show the letters A, B, C, D, E, A. The die is thrown once. Find the probability of getting **3**

(i) A

(ii) D

(iii) a vowel.

OR

A lot of 20 bulbs contains 4 defective ones. One bulb is drawn at random. **3**

(i) Find the probability that it is defective.

(ii) If the bulb drawn is not defective and is not replaced, find the probability that the next bulb drawn is defective.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting **5**

(i) a spade

(ii) the queen of diamonds

(iii) a red ace

(iv) neither a heart nor a king

(v) a black card or a king.

OR

Two dice are thrown together. Find the probability that **5**

(i) the sum is 9

(ii) the sum is a multiple of 3

(iii) the difference of the numbers is 2

(iv) the sum is at least 10.

20. **Case study:** At a school fair, a bag has tokens numbered 1 to 30. A player wins if the token drawn is a multiple of 5 or a perfect square. **5**

(i) Find P(multiple of 5).

(ii) Find P(perfect square).

(iii) Find P(winning).

(iv) A student says P(winning) $= \frac{11}{30}$. Find the mistake.

OR

A bag contains 18 balls out of which x are red. **5**

(i) If one ball is drawn at random, find the probability that it is not red.

(ii) If 2 more red balls are put in the bag, the probability of drawing a red ball becomes $\frac{9}{8}$ times the earlier probability. Find x.

END OF QUESTION PAPER — Total: 40 Marks

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